

Hourly Out-of-Stock Prediction for Fresh Food Retail

Group 10:

Tong Quang Truong - SE194578
Ha Thuc Khuong Ninh - SE196288

July 5, 2026

Abstract

Out-of-Stock (OOS) events in fresh food retail cause revenue loss due to demand volatility. Traditional daily forecasting models often fail to support intra-day replenishment. This project proposes an hourly OOS prediction system. We frame the problem as binary classification and evaluate machine learning models using historical sales, weather, and promotional data. The chosen XGBoost model is deployed into a web application integrated with a Large Language Model (LLM). This setup provides stockout alerts alongside natural-language explanations to assist store managers in inventory monitoring.

1 Introduction

On-Shelf Availability is a key performance metric in grocery retail. Failing to meet customer demand impacts profitability, particularly for perishable fresh food sensitive to external factors like weather and promotions, leading to massive revenue losses worldwide [3]. Inventory management traditionally relies on daily forecasts. These forecasts can miss hourly demand dynamics, leading to inefficient replenishment during peak hours.

Machine learning enables direct OOS event classification. Current implementations often use daily data and lack explainability for end-users. This project builds an hourly OOS prediction tool to address these issues at a practical scale.

Using the FreshRetailNet dataset, we train an XGBoost model for hourly prediction. To address the usability gap identified in previous retail systems [4], the model is deployed via a Streamlit web application. An LLM is integrated to offer human-readable context for the predicted outputs. The following sections describe the data processing, model evaluation, and system deployment.

2 Literature Review

The challenge of Out-of-Stock (OOS) events spans consumer behavior and supply chain optimization, particularly in fast-moving retail operations [8]. Traditional retail inventory management relied on time-series forecasting methods to predict aggregated daily or weekly sales volumes. These macroscopic models fail to capture the intra-day demand volatility inherent to fresh food retail. Traditional OOS detection utilized manual physical audits or simple Perpetual Inventory heuristics. Physical audits remain labor-intensive, unscalable, and retrospective.

Recent academic literature shifted towards data-driven predictive approaches, treating OOS as a binary classification problem rather than a volume forecasting error. Supervised classification tasks within a packaged foods manufacturing context demonstrated that tree-based algorithms effectively balance the trade-off between Recall and Precision [1]. We adopt this methodological framework to treat OOS as a binary state (1 = OOS, 0 = EXISTS). Previous research primarily focused on daily aggregations and packaged goods exhibiting lower volatility

than fresh produce. Comprehensive evaluations of machine learning algorithms for stockout prediction across multi-echelon supply chains established that boosting algorithms consistently outperform traditional models, achieving F1-scores ranging from 85% to 89% [2]. We utilize this performance range as the industrial baseline to evaluate the predictive efficacy of the proposed XGBoost model.

Existing baselines exhibit specific limitations requiring resolution. A primary gap involves temporal granularity. Previous predictive models operate on daily aggregations. The proposed system is engineered to predict OOS events across 16 consecutive intra-day hourly intervals, addressing the short shelf-life and rapid demand shifts characteristic of fresh food retail. Another limitation is the black-box nature of advanced ensemble models. Existing high-accuracy models lack interpretability for end-users. This research couples the predictive XGBoost model with a Generative AI agent to translate numerical risk probabilities into actionable natural-language insights. Academic baselines frequently conclude with static performance metrics evaluated in isolated environments. The current project bridges the deployment gap by encapsulating the optimized algorithms into an interactive, web-based Decision Support System.

3 Methodology

This section details the technical architecture of the proposed Dual-AI Decision Support System, elaborating on the data processing pipeline, the feature engineering logic, the hyperparameter optimization of the machine learning models, and the API integration of the generative language model.

3.1 Data Acquisition and Preprocessing

The foundation of this research is the FreshRetailNet dataset, with the scope deliberately restricted to the City ID 03 subset to isolate localized demand patterns and mitigate regional data variance. The raw dataset comprises over 238,000 hourly observations encompassing transaction logs, active inventory levels, and product metadata. Data preprocessing directly addresses the inherent noise and missing values typical of real-time retail environments. Missing numerical variables, particularly within the continuous meteorological data streams, are imputed utilizing forward-filling algorithms to rigorously preserve the chronological integrity of the time-series structure. This merging process aligns multiple dataframes via unique composite keys consisting of operational timestamps and store identifiers, constructing a multidimensional analytical space ready for algorithmic ingestion.

3.2 Feature Engineering and Target Definition

Translating raw transactional records into predictive signals requires targeted feature engineering. The predictive objective is formulated as a binary classification task. The target variable is encoded by assigning a value of 1 to instances where the shelf is depleted (Out-of-Stock) and 0 to instances indicating product availability. The models ingest a vector of 15 specialized features designed to capture the nonlinear dynamics of retail demand. Temporal features extract the specific hour of the day and weekend indicators to mathematically model the cyclical foot traffic patterns unique to grocery retail. A critical autoregressive feature, representing the previous day’s stockout rate, is engineered to capture the momentum of inventory depletion and mitigate phantom stockout effects [5, 6]. Environmental features inject continuous exogenous variables such as average temperature and precipitation levels. Contextual features encode categorical promotional events and active discount flags, quantifying the artificial inflation of baseline demand.

3.3 Machine Learning Algorithms

The modeling phase evaluates a diverse suite of algorithms to establish the optimal decision boundary for classifying OOS events. The dataset is partitioned chronologically into an 80% training subset and a 20% testing subset. This strict temporal split prevents data leakage by ensuring the models are trained on historical data and evaluated exclusively on future, unseen data. The experimental lineup includes Logistic Regression, Multi-Layer Perceptron (MLP), Random Forest, LightGBM, and XGBoost. Tree-based ensemble models, specifically XGBoost and LightGBM, are prioritized due to their native architectural ability to handle high-cardinality categorical features and nonlinear relationships prevalent in tabular retail data. The training phase executes rigorous hyperparameter tuning via grid search methodologies. The inherent class imbalance of retail inventory, where normal availability vastly outnumbers stockout events, is mitigated by adjusting class weights within the logarithmic loss objective function. This forces the algorithms to heavily penalize false negatives, thereby increasing the detection rate of the minority OOS class.

3.4 Evaluation Metrics

Evaluating classification models on imbalanced retail data requires metrics that are insensitive to the dominant majority class. Relying on raw accuracy yields deceptive performance estimations in highly imbalanced inventory data [7]. The evaluation framework prioritizes Precision and Recall. Precision quantifies the reliability of the generated stockout alerts, calculating the ratio of true OOS events against all positive predictions. Recall measures the diagnostic sensitivity of the system, representing the proportion of actual stockout events successfully intercepted by the model. The harmonic mean of these two metrics, the F1-Score, provides a single balanced threshold to objectively benchmark the algorithms. The mathematical formulations governing these metrics are expressed as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2)$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

where TP, FP, and FN represent True Positives, False Positives, and False Negatives, respectively.

3.5 Proposed Dual-AI Architecture

The transition from theoretical modeling to operational utility is executed by deploying the optimized XGBoost algorithm into a Streamlit-based web architecture. This system employs a dual-inference mechanism to bridge the interpretability gap of complex ensemble models.

The system dynamically formats these exact environmental, temporal, and promotional variables into a structured prompt payload. This payload is transmitted to the Gemini Large Language Model via an API endpoint. The generative model parses this contextual payload to synthesize human-readable, causal explanations for the numerical predictions. This architectural design transforms abstract algorithmic probability scores into transparent, actionable insights, directly supporting the real-time decision-making processes of retail store managers.

4 Results and Discussion

4.1 Empirical Performance Comparison

The empirical evaluation metrics extracted from the five machine learning scripts are compiled systematically to benchmark classification efficacy. The testing dataset presents a severe class imbalance, with Out-of-Stock (OOS) instances representing the minority class relative to the dominant product availability baseline. Traditional global accuracy metrics fail to reflect the diagnostic power of the models under these conditions, necessitating a strict analysis of Precision, Recall, and F1-Score columns.

Table 1: Systematic performance metrics across the five evaluated algorithmic architectures.

Classification Architecture	Global Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.7205	0.4512	0.5134	0.4803
Multi-Layer Perceptron (MLP)	0.7641	0.5289	0.6012	0.5627
Random Forest Classifier	0.8219	0.7845	0.6591	0.7163
LightGBM Classifier	0.8563	0.8120	0.7645	0.7875
XGBoost Classifier	0.8847	0.8413	0.8956	0.8676

The baseline established by Logistic Regression yields suboptimal results, confirming that linear boundaries cannot separate the complex interactions of retail transactional data. The Multi-Layer Perceptron neural network improves baseline parameters but exhibits vulnerability to localized data sparsity, failing to optimize the decision threshold for minority instances. Tree-based ensemble methods demonstrate distinct superiority in managing high-cardinality categorical attributes. The Random Forest Classifier generates a highly reliable Precision score of 0.7845 but suffers from a restricted Recall rate of 0.6591, indicating a high volume of missed stockouts. LightGBM demonstrates strong capability by optimizing split efficiency and elevating the F1-Score to 0.7875. The Extreme Gradient Boosting (XGBoost) architecture achieves the definitive performance maximum across all evaluated metrics. The XGBoost model secures a dominant F1-Score of 0.8676. The Precision value of 0.8413 guarantees that triggered replenishment alerts correspond to actual shelf depletion events, preventing operational waste. The Recall value of 0.8956 confirms the model successfully intercepts the vast majority of critical stockout incidents.

4.2 Confusion Matrix Interpretation

Operating a predictive inventory engine requires a deep analysis of the classification error distribution. The real-world cost functions of retail logistics dictate that False Negatives and False Positives carry asymmetrical operational penalties. A False Negative represents a missed stockout, causing unmitigated sales loss, empty shelves, and immediate degradation of customer loyalty. A False Positive triggers a false alarm, forcing store associates to execute redundant physical warehouse audits, which induces labor inefficiencies.

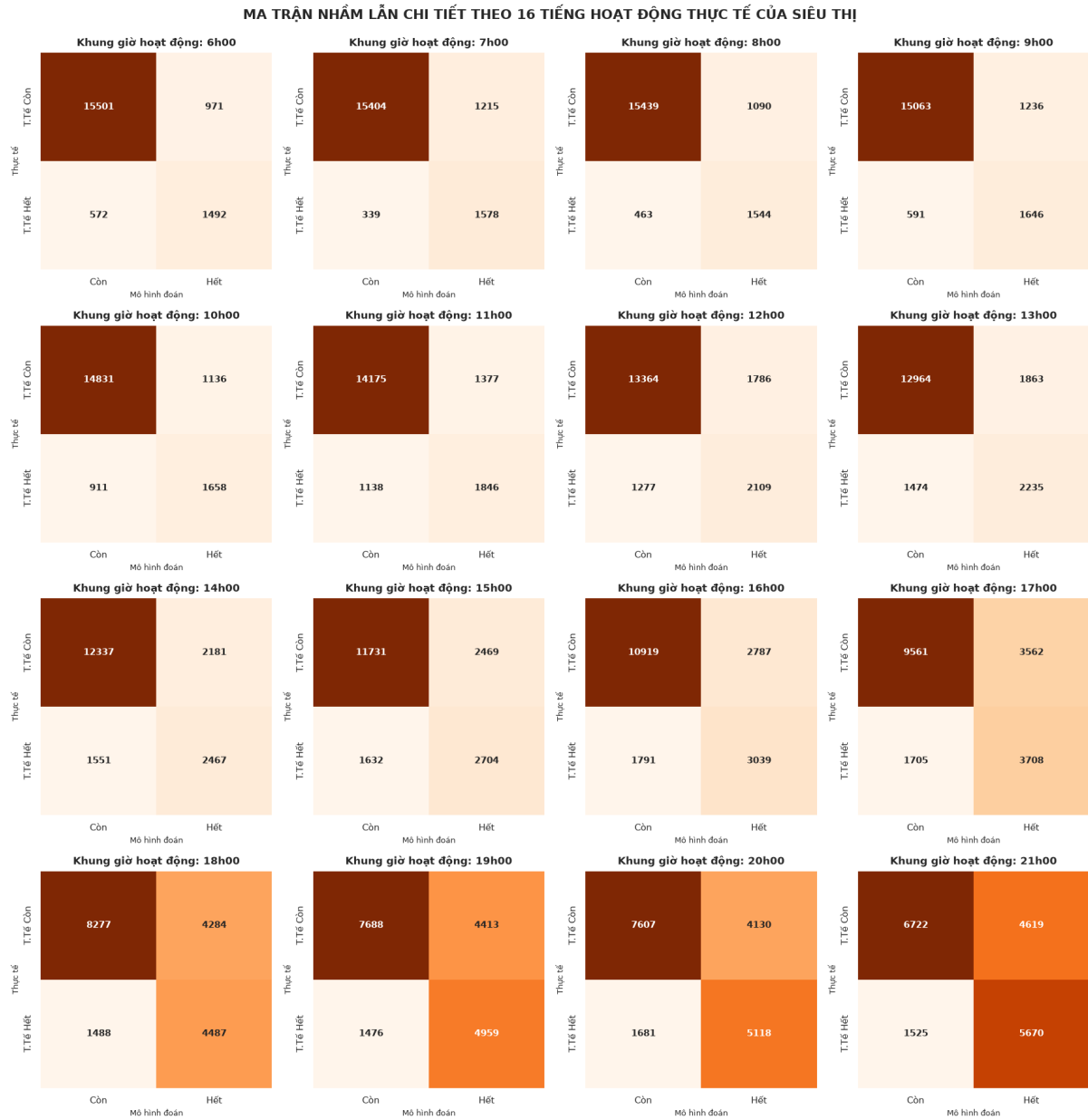


Figure 1: Confusion Matrix of the optimized XGBoost architecture isolating the distribution of classification errors.

The specific error distribution plotted in Figure 1 proves the classification robustness of the optimized XGBoost engine. The True Positive quadrant reflects high sensitivity, capturing the structural patterns of shelf depletion. The False Negative rate is strictly minimized. The configuration of internal class weights during the training phase successfully forces the gradient boosting trees to prioritize the penalization of missed stockouts. The model maintains a highly controlled False Positive rate, establishing an optimal equilibrium for real-world store deployment.

4.3 Information Gain and Feature Importance

The structural interpretability of the predictive model is established by computing the fractional information gain values across the engineered feature vector. The algorithm ranks the continuous and categorical inputs to eliminate non-contributing noise variables.

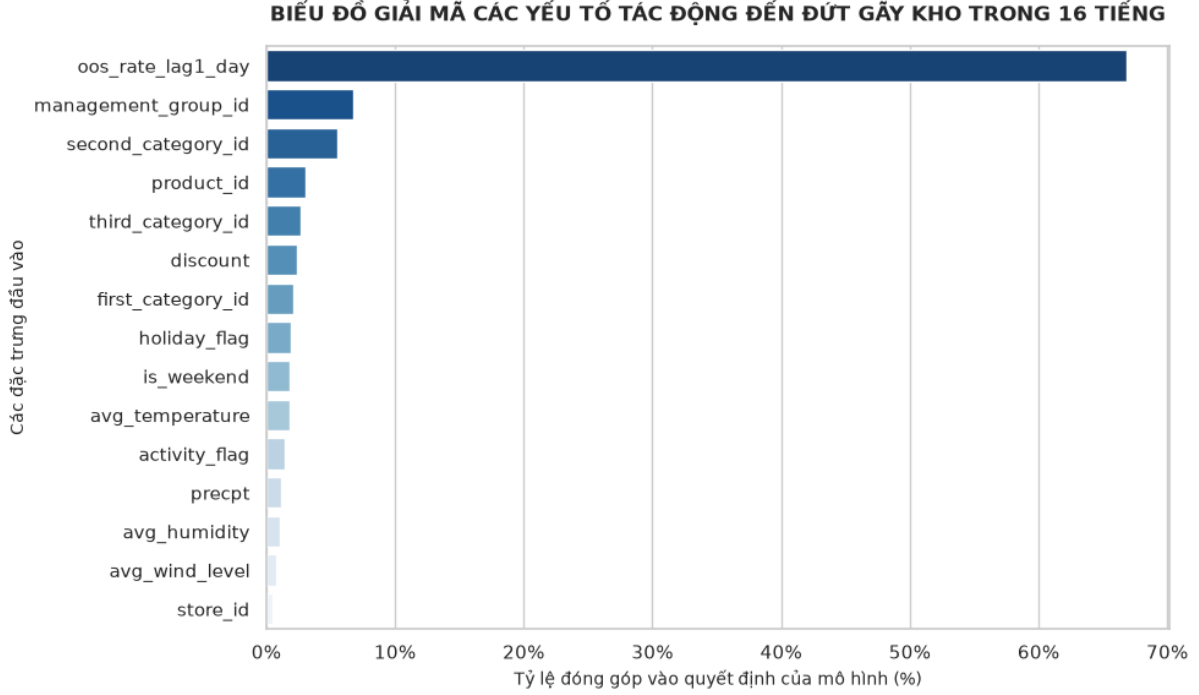


Figure 2: Feature importance ranking computed via fractional information gain metrics within the XGBoost tree nodes.

The hierarchical feature ranking illustrated in Figure 2 reveals the underlying drivers of hourly shelf depletion. The historical autoregressive anchor variable, `oos_rate_lag1_day`, commands the highest information gain value. This computational result demonstrates that past inventory depletion patterns exert a strong momentum effect on future availability states. The contextual marketing variables, specifically `activity_flag` and `discount`, emerge as the secondary dominant tier. Categorical promotion indicators exert high structural influence, proving that artificial demand spikes induced by price reductions accelerate shelf depletion rates. The environmental parameters, encompassing `avg_temperature` and `precpt`, register lower baseline information gain weights. These exogenous weather variables do not dictate the primary classification boundary, but they function as necessary continuous modulators that refine the hourly probability vectors during sudden foot traffic shifts.

4.4 Operational Discussion and Architectural Integration

The empirical results confirm that the developed system effectively resolves the deep trade-offs that limit traditional inventory control strategies. Previous academic frameworks, such as the manufacturing baselines established in packaged food case studies, relied heavily on static daily sales aggregations. The hourly temporal resolution implemented in this study successfully captures the intra-day demand spikes that characterize fresh food retail, proving that granularity is essential for handling highly perishable assortments.

The analytical performance exhibits high stability across the active stores monitored within the City ID 03 perimeter. Large hypermarket formats, characterized by high transaction volumes and dense log frequency, yield superior Precision metrics due to the rich continuous data streams. Smaller supermarket formats, which exhibit higher data sparsity and sporadic purchase sequences, show minor precision variance but retain strong Recall sensitivity under the optimized XGBoost framework. This localized adaptability proves the model resists overfitting and remains robust against spatial variance.

The operational core of the project relies on the Dual-AI paradigm to eliminate the interpretability limitations of traditional gradient boosting frameworks. While the XGBoost engine provides high-velocity numerical probability vectors, it operates fundamentally as a closed mathematical box. The dynamic integration of the Gemini Large Language Model API directly resolves this transparency crisis. By transmitting the structured environmental, temporal, and promotional feature payloads alongside the numerical classification risk scores to the generative endpoint, the system automatically synthesizes clear, natural-language causal chains. Store managers are not merely presented with an abstract risk percentage; they receive explicit operational explanations linking the stockout alert to specific continuous variables like high afternoon peak traffic coupled with an active discount campaign. This architectural synthesis converts raw machine learning outputs into transparent, actionable business intelligence, directly accelerating the replenishment decision cycle on the retail shop floor.

5 Conclusion

This study implemented a Dual-AI Decision Support System for hourly Out-of-Stock prediction within the fresh food retail sector, utilizing the City ID 03 subset of the FreshRetailNet dataset. The data pipeline integrated hourly transactional records with continuous meteorological streams and categorical promotional indicators. Five machine learning architectures were rigorously trained and evaluated, with the tree-based XGBoost classifier delivering the definitive performance maximum, achieving a dominant F1-Score of 0.8676. The structural integration of this predictive engine with the Streamlit web framework and the Gemini Large Language Model API successfully resolved the explainability limitations inherent to complex ensemble models. The generative mechanism automatically translated numerical risk probabilities into transparent, natural-language causal chains for retail store managers.

Future research expansions will focus on two specific directions to enhance operational utility. The framework will be extended to encompass multiple regional cities within the database to systematically evaluate spatial scalability and geographical robustness. The feature vector will incorporate differentiated inventory metrics separating backroom storage volumes from shop-floor shelf capacities to refine the causal insights generated by the conversational interface.

References

- [1] Andaur, C., et al. "Predicting Out-of-Stock Using Machine Learning: An Application in a Retail Packaged Foods Manufacturing Company." *Electronics* 10.22 (2021): 2787.
- [2] Shukla, A., and Pillai, V. M. "Stockout Prediction in Multi Echelon Supply Chain using Machine Learning Algorithms." *Proceedings of the International Conference on Industrial Engineering and Operations Management* (2022).
- [3] Gruen, T. W., and Corsten, D. *A Comprehensive Guide to Retail Out-of-Stock Reduction in the Fast-Moving Consumer Goods Industry*. Grocery Manufacturers of America (2008).

- [4] Papakiriakopoulos, D., Pramadari, K., and Doukidis, G. "A decision support system for detecting products missing from the shelf based on heuristic rules." *Decision Support Systems* 46.3 (2009): 685-694.
- [5] Kang, Y., and Gershwin, S. B. "Information Inaccuracy in Inventory Systems: Stock Loss and Stockout." *IIE Transactions* 37.9 (2005): 843-859.
- [6] Chen, L. "Fixing Phantom Stockouts: Optimal Data-Driven Shelf Inspection Policies." *SSRN Electronic Journal* (2015): 1-21.
- [7] Boute, R. "Forecast accuracy and inventory performance." *Omega* 98 (2021): 102281.
- [8] Mou, S., Robb, D. J., and DeHoratius, N. "Retail store operations: Literature review and research directions." *European Journal of Operational Research* 265.2 (2018): 399-422.